Tasneea Zahra

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Education

Khulna University of Engineering & Technology (KUET) *Jan 2020 – Sept 2025*
Bachelor of Science in Electrical and Electronic Engineering; Major: CSP

- **CGPA:** 3.61/4.0
- **Relevant Coursework:** Digital Image Processing, Microprocessors, Numerical Methods and Statistics, Data Structures and Algorithms, Wireless Communication, Power Electronics and Industrial Drives, VLSI Design and Technology, Electrical Machines, Control Systems, Digital Electronics and Logic Design, Communication Engineering, and Digital Signal Processing.

Thesis: Implementation and Performance Analysis of a Hybrid CNN–Transformer Model for Polyp Image Segmentation.

Research Focus

My research interests span machine learning, computer vision, image and signal processing, and natural language processing. I am especially interested in medical image segmentation and vision-language question answering, as well as retrieval-augmented systems for research discovery and efficient model fine-tuning.

Publications & Preprints

[1] **Skip What You Can Predict: Predictive Repositioning for Policy Optimization for Efficient LLM Training**
I. N. Swapnil, A. Saha, **T. Zahra**, T. A. Khan, M. A. Haque, and S.-N. Lim.
Rather than executing every same-batch optimizer step in RLVR, the method extrapolates from two observed steps to a farther point on the trajectory and corrects it. It matches performance with fewer optimization steps and less wall-clock time, with error bounds for AdamW and Muon.
Submitted to ICLR 2027; Preprint; arXiv:2605.06755.

[2] **Trans-SPAnet: A Unified Transformer–CNN Architecture with Enhanced Boundary Optimization for Polyp Segmentation**
IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON 2025).

Research Experience

University of Central Florida *Nov 2025 – Present*
Research Collaborator *Orlando, FL (Remote)*

- Collaborating with Prof. Ser-Nam Lim on efficient policy optimization for large language model reasoning.
- Co-author of PrePO, which uses predictive repositioning to approximate repeated same-batch optimizer updates at a fixed active cost, reducing explicit optimization steps and wall-clock time at matched performance.
- Established finite-horizon error bounds for AdamW and Muon; validated the method on reinforcement learning with verifiable rewards for language-model reasoning, with gains carrying over to supervised fine-tuning.

Independent Research
Medical AI and Research Literature Question Answering

- Built and evaluated a medical visual question-answering pipeline for pathology images, fine-tuning a vision-language model with Transformers and PEFT-based LoRA/QLoRA.
- Built a PDF-based research assistant using Gemini File Search for grounded paper exploration, summarization, and citation lookup across uploaded papers.

Khulna University of Engineering & Technology (KUET)
Undergraduate Thesis Research

- Designed and implemented a hybrid CNN–Transformer architecture for 2D polyp image segmentation.
- Combined spatial-context preservation blocks with a Transformer-based decoder and analyzed segmentation performance.

Selected Projects

- **Bangla OCR Pipeline**: *Jun 2025 – Present; Dhaka, Bangladesh.* Combined YOLO word detection with Qwen3-VL-2B recognition for handwritten and printed Bangla, with PyTorch and ONNX deployment backends.

- **Implementation and Performance Analysis of a Hybrid CNN–Transformer Model for Polyp Image Segmentation**:
  Designed and implemented a hybrid CNN–Transformer architecture for 2D polyp image segmentation. Combined spatial-context preservation blocks with a Transformer-based decoder to improve segmentation performance.
- **Research Paper Question-Answering Chatbot with RAG and LLM Orchestration**:
  Built a PDF-based research assistant for paper exploration, summarization, and citation lookup using Gemini File Search. It grounds responses in text, tables, and figures from uploaded papers and supports multi-session chat and document libraries.
- **Medical Visual Question Answering for Image-Based Clinical Understanding**:
  Built and evaluated a medical visual question-answering pipeline that answers natural-language questions about pathology images. Applied efficient fine-tuning to a vision-language model using Transformers, PEFT-based LoRA/QLoRA, and Unsloth-compatible libraries.

Technical Skills

- **Programming and Data Analysis**: Python, MATLAB, Pandas, NumPy
- **Machine Learning and Deep Learning**: PyTorch, Transformers
- **LLMs and Generative AI**: Fine-tuning, Unsloth, retrieval-augmented generation (RAG), LangChain, LangGraph, AI agents
- **Circuit Design**: PSpice, LTspice, Proteus, LOGO! Soft Comfort, Altera, Quartus, Arduino IDE
- **Productivity and Documentation**: Microsoft Word, Excel, PowerPoint, LaTeX

Industrial Visits

- Karnaphuli Hydropower Station
- Cox's Bazar Cable Landing Station
- Radar Station, Cox's Bazar

Honors and Achievements

- **Dean's List Award**, KUET — Recognized for earning a GPA of 3.75 or higher during the 2019–2020 academic session.
- **Technical Education Scholarship** — Recognized for ranking among the top 60 students in class.
- **Machine Learning with Python** — Completed an intermediate course through the Edge University Activation Program.

Extracurricular Activities

- Owner of **Crochet of Love**, an online business specializing in handmade crafts.
- General member of EEE Makers Hub and KUET Debating Society.
- Presidium member of the RAJUKians of KUET (ROK) Association.
- Organized EEE Day 2023.
- Hosted several university events.
- Performed in university cultural programs.

References

Dr. Kalyan Kumar Halder
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